

Building the software stack of the FT era

QRE Workshop

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Q-CTRL

Closing development gaps

Software for FTQC continues to trail hardware progress

Challenge:

Beyond hardware development, current FTQC research remains largely theoretical and top-down, often overlooking the practical architectural barriers of future fault-tolerant systems.

Objective:

Building a modular software architecture for fault-tolerant execution that delivers:

- Precise resource estimates
- Informs hardware and algorithm co-design
- Accelerates quantum error correction roadmaps
- Provides the backbone of the future full stack

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Components of the computing stack

Modular design of infrastructure software



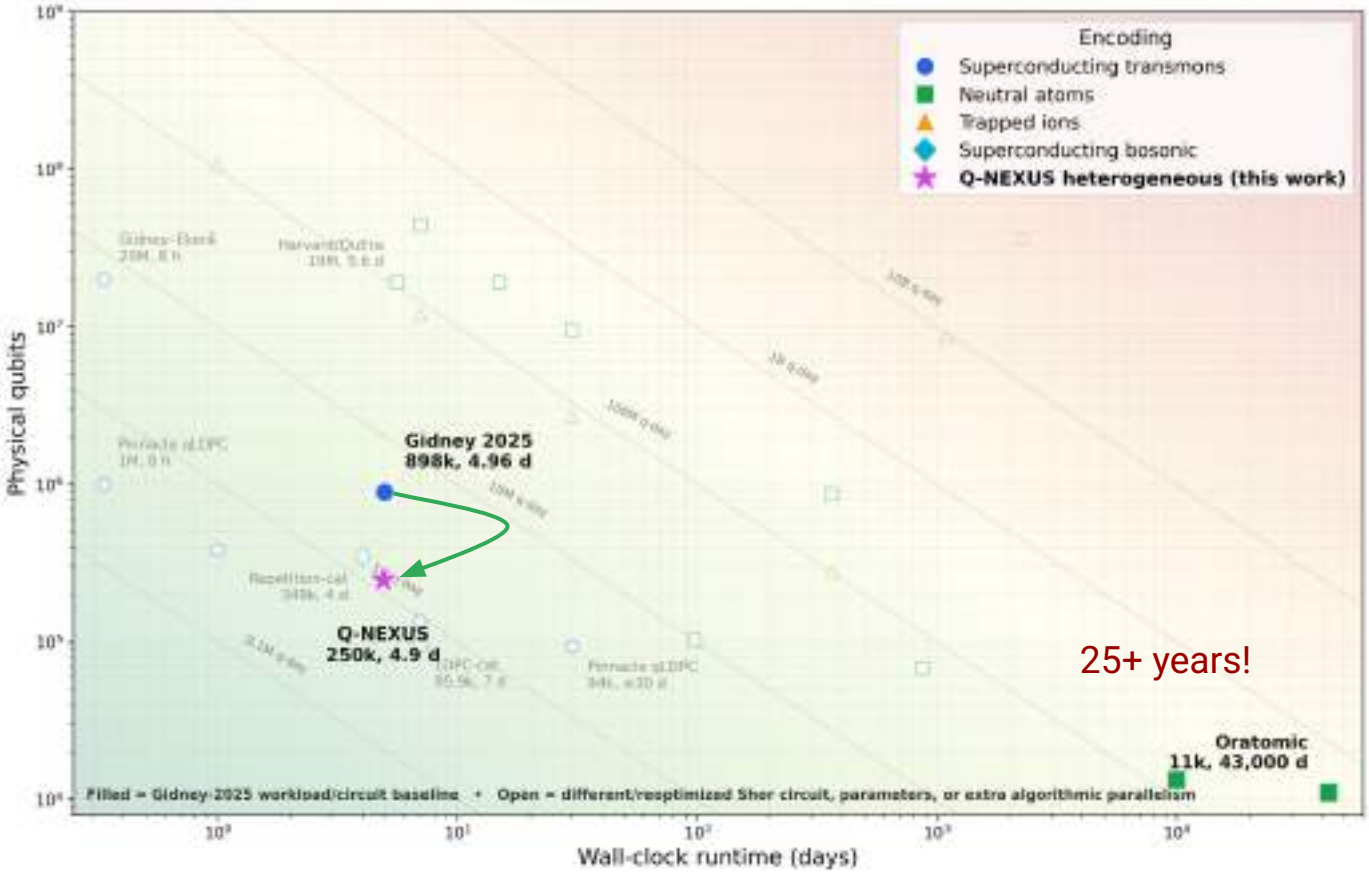
Q-CTRL is systematically advancing every layer of the quantum software stack, already demonstrated by several key milestones:

- **Modular architecture execution**: Demonstrated optimal execution across modular systems.
- **Advanced compilation**: Developed a state-of-the-art Clifford+T synthesizer that outperforms existing compilation tools.
- **Hardware-aware resource estimation**: Built a resource estimation tool that accounts for physical machine constraints using exact scheduling and optimal compilation.

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Moving the pareto front of FTQC resources is not only about top-down QEC code exploration

Bottom-up approaches for reducing space-time overheads

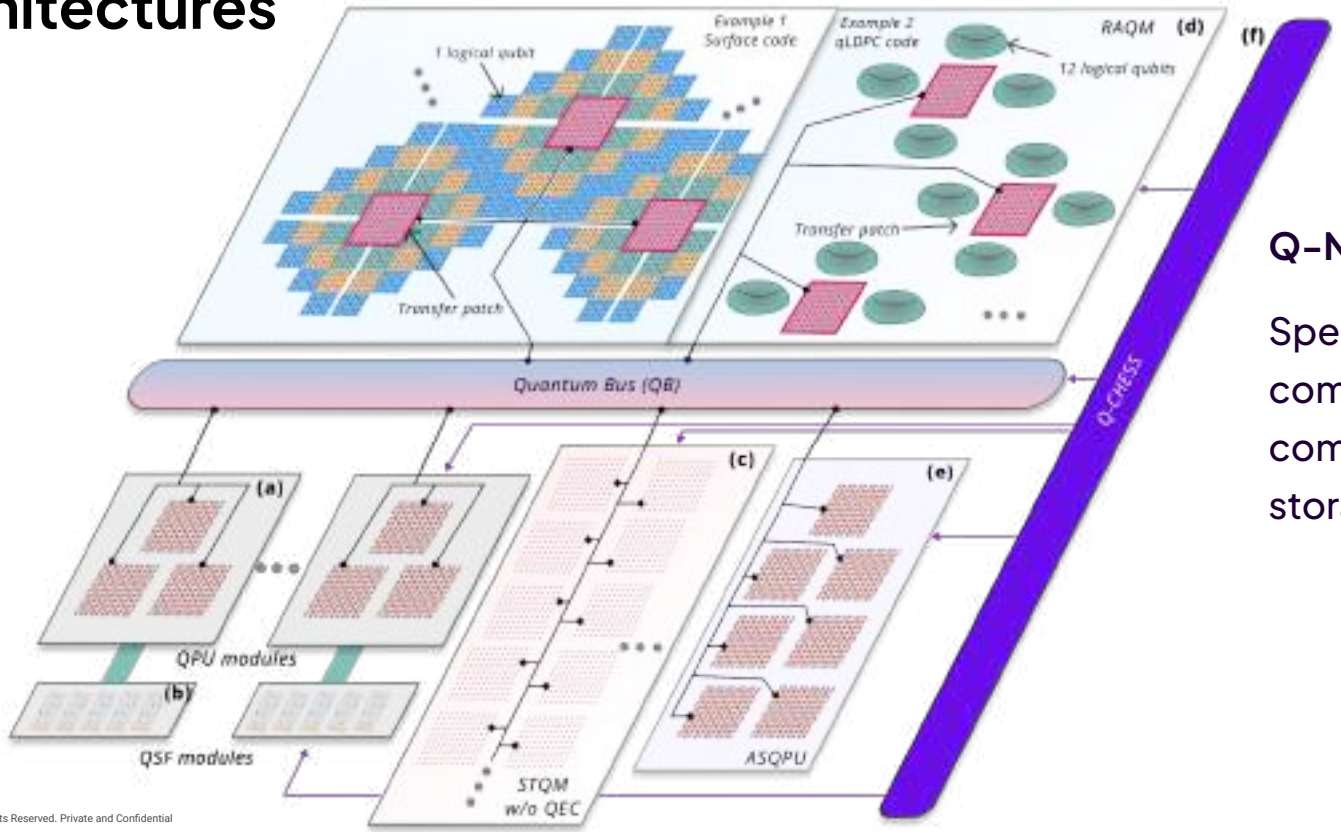


[1] Gidney- Ekera arXiv:1905.09749
[2] Gouzien et al. arXiv:2302.06639
[3] Ruis et al. arXiv:2401.09541
[4] Zhou et al. arXiv:2505.15907
[5] Gidney arXiv:2505.15917
[6] Webster et al. arXiv:2602.11457
[7] Cain et al. arXiv:2603.28627
[8] Mundada et al. arXiv:2604.06319

Fault tolerant architectures

Moving away from hand designed architectures

Q-NEXUS: Software for defining and handling arbitrary architectures



Q-NEXUS:

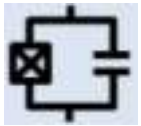
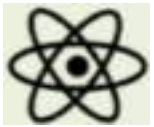
Specialized modules for computation, communication, and storage.

arXiv:2604.06319

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Pushing heterogeneity to the edge

Matching modality to functionality

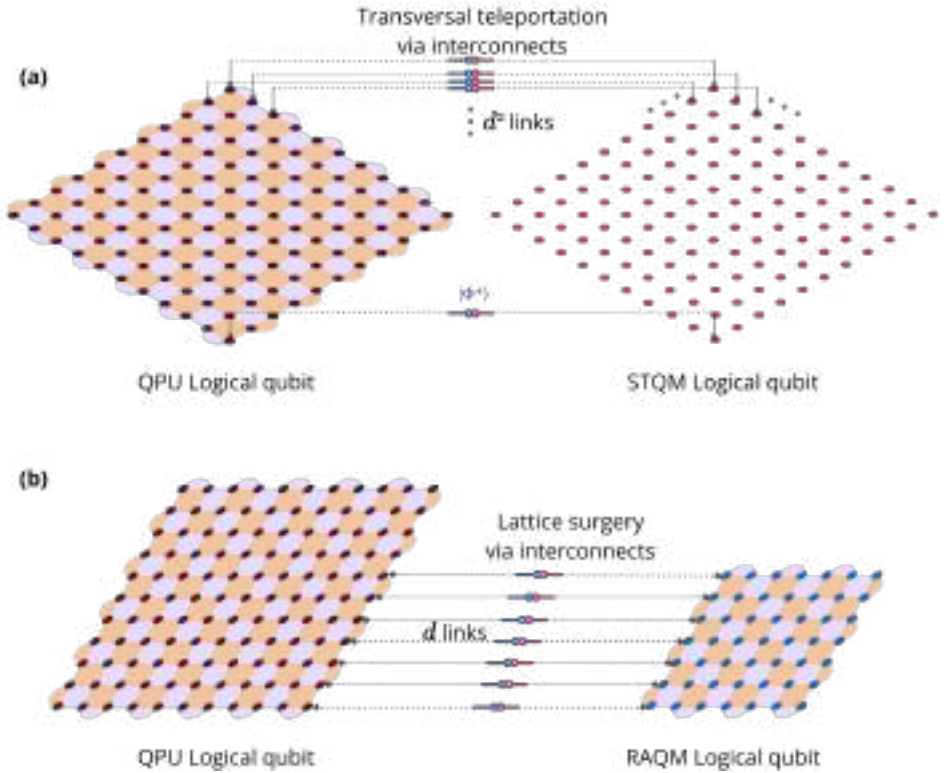
				
Long-range connectivity	❌	✅	✅	✅
Speed	✅	❌	❌	✅
Coherence	❌	✅	✅	❌
High fidelity gates	✅	✅	✅	❌

...

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Interconnects are part of the computation and their resources can't be ignored

Protocols for logical state transfer



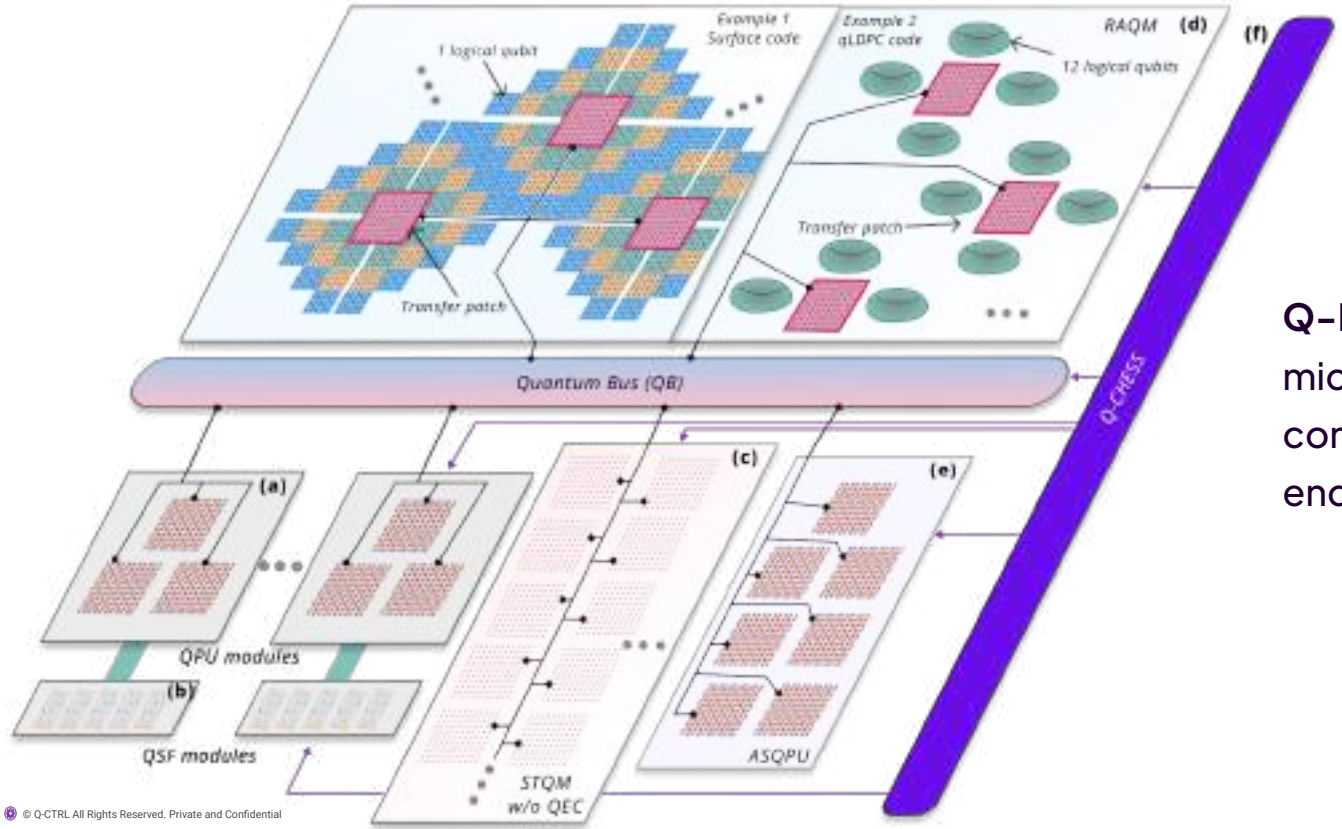
State transfer protocol:

1. **Prepare rails**: generate and purify long-range physical Bell pairs
2. **Teleport:**
   a. Transversal: consume Bell pairs to teleport each physical data qubit
   b. Lattice surgery: consume Bell pairs at the start of QM clock cycle to perform lattice surgery (XX/ZZ checks) over d_{QPU} QEC rounds in QM and update pauli frame corrections

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Realistic resource estimation: going beyond counting T-gates

Q-CHESS: end-to-end orchestration



Q-NEXUS is integrated with a microarchitecture-aware compiler (**Q-CHESS**) for end-to-end orchestration.

arXiv:2604.06319

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Compiler and scheduler

Facing HW constraints and realities in a systematic way

Fully functional compiler and scheduler for FTQC pipelines

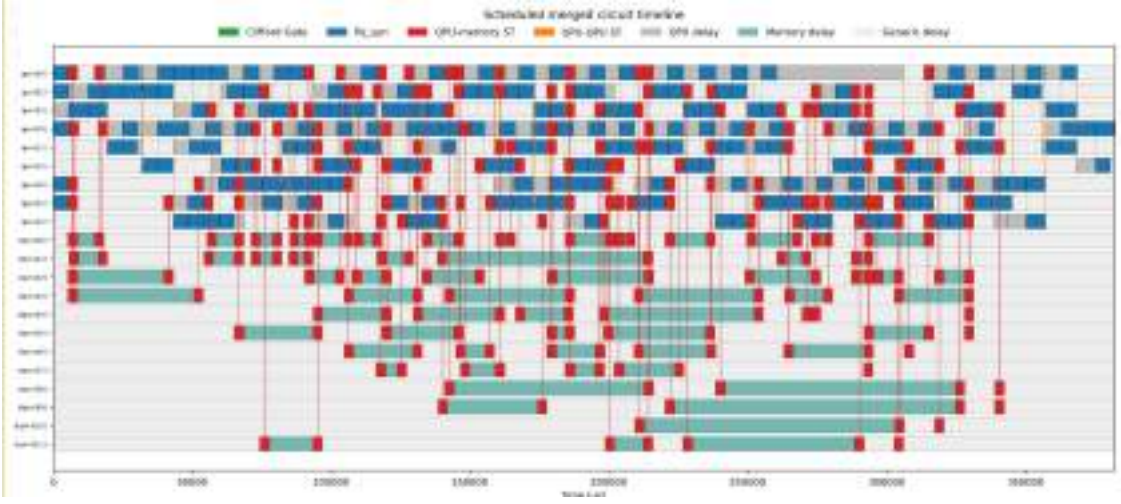
User Input

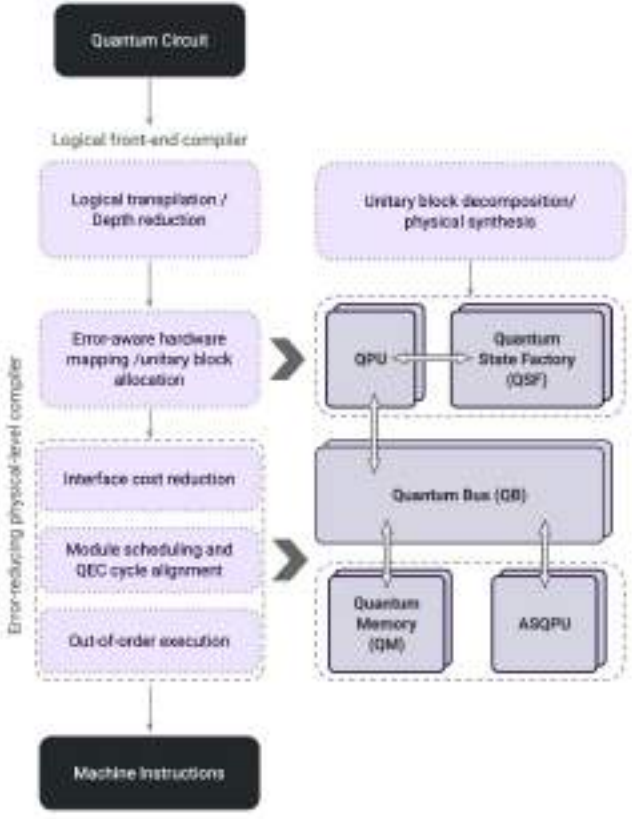
- Representative parameters e.g. coherence time, gate fidelity, code rate, connectivity, communication latencies
- Arbitrary abstract quantum program

Q-CHESS
Fully-functional software for heterogeneous compilation

Output

- Resource estimation Execution time, circuit fidelity, operations per qubit species, qubit number overhead
- Physically implementable circuit



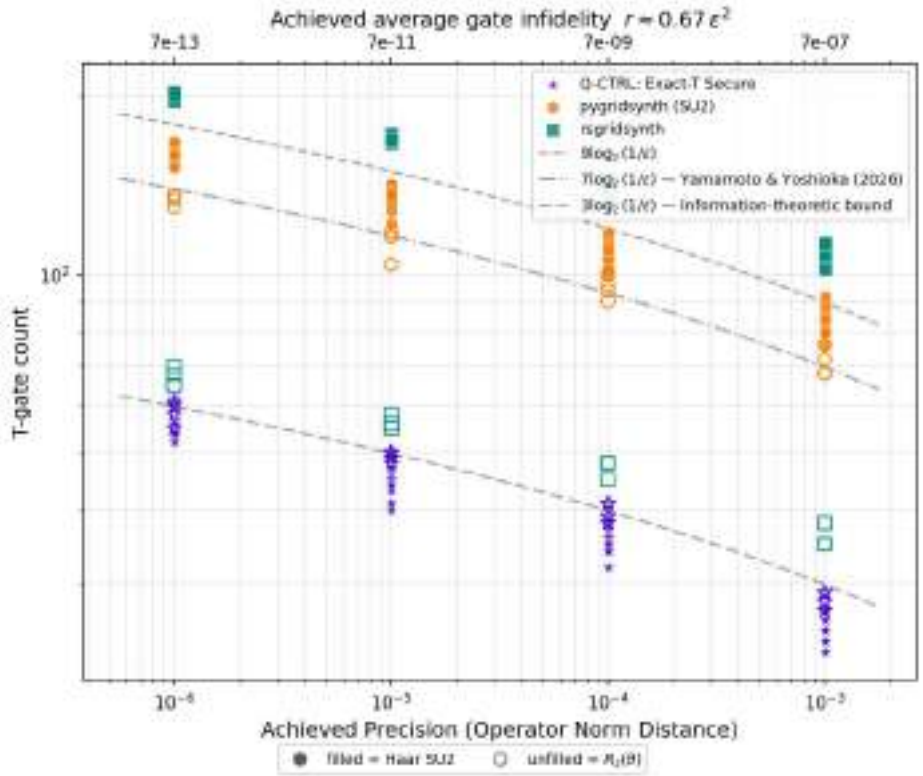


arXiv:2604.06319

Compiler and scheduler

Provably optimal Clifford + T synthesis

Suboptimal / naive pipelines lead to wrong resource and error estimation



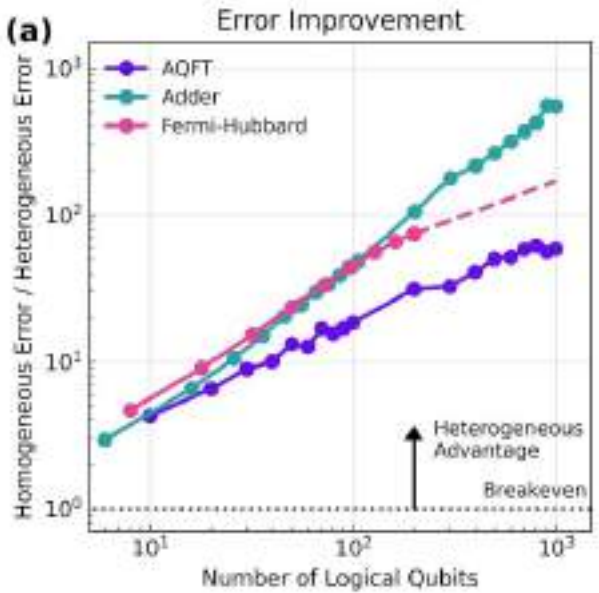
To be released soon

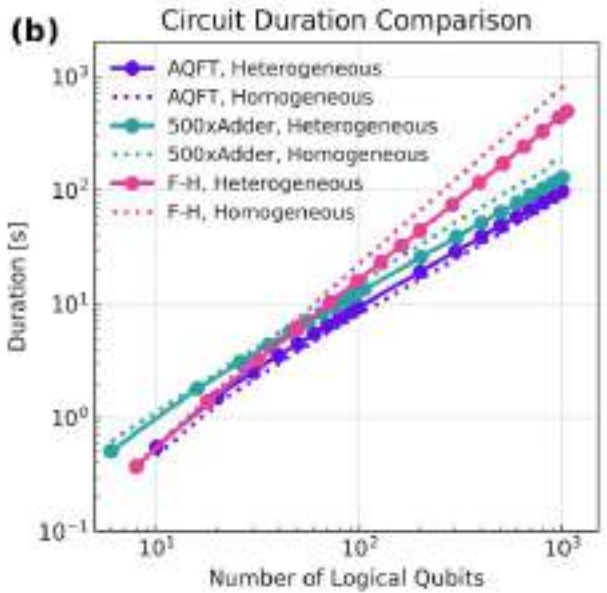
- Provably lowest T-count synthesizer
- Highly performant standalone package that can be easily integrated with any FTQC evaluation pipeline

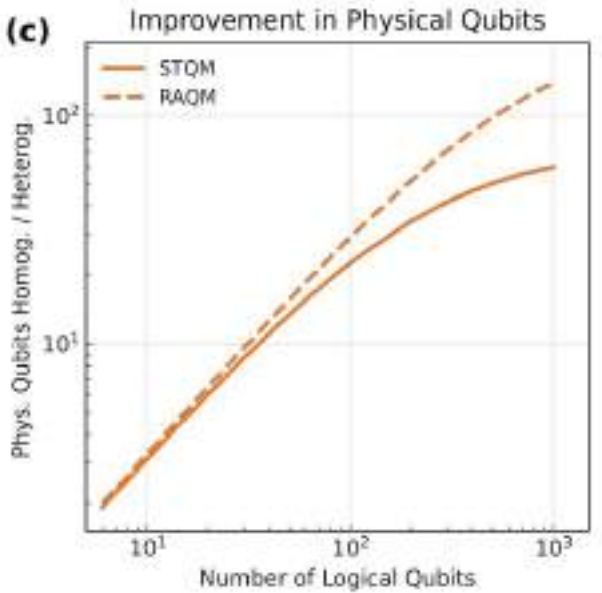
Comparing to monolithic architectures

Resource estimator

Orders of magnitude improvements across various key applications





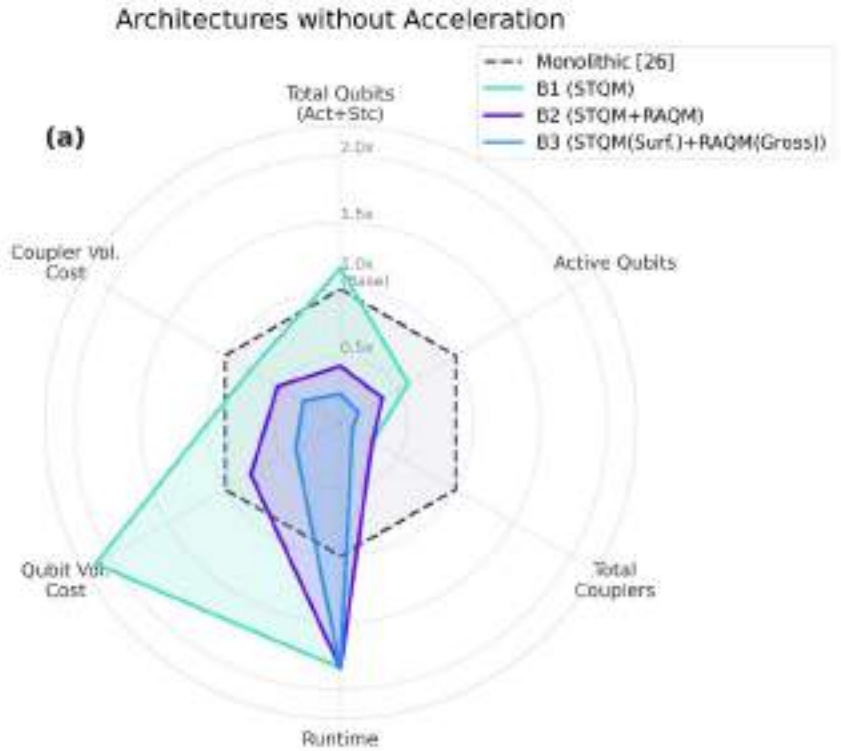


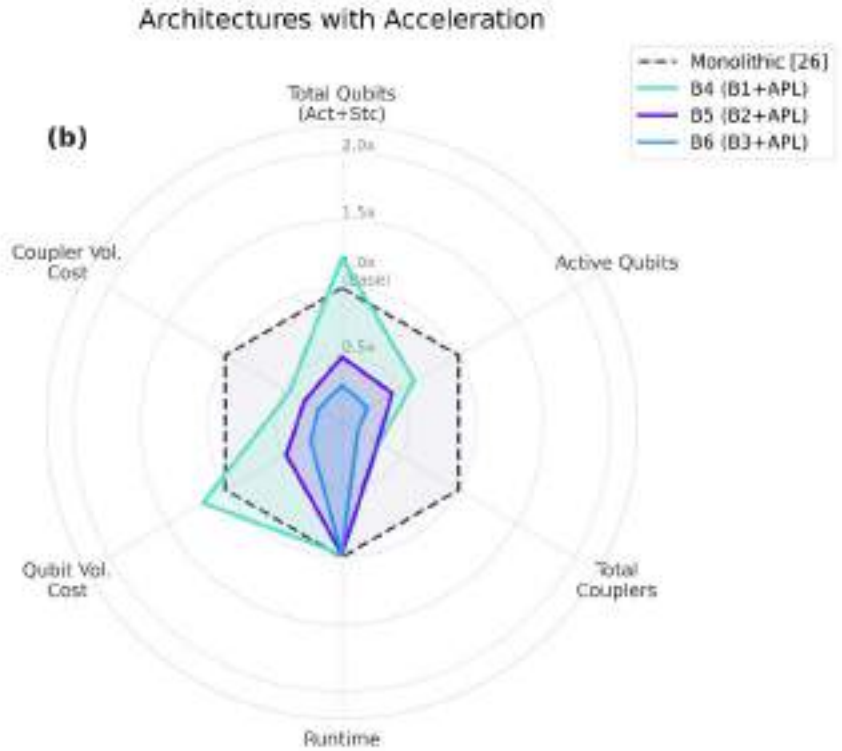
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Realistic error estimates – comparing to hand designed architectures

Resource estimator

Improving the space–time volume of RSA-2048 with no hidden cost





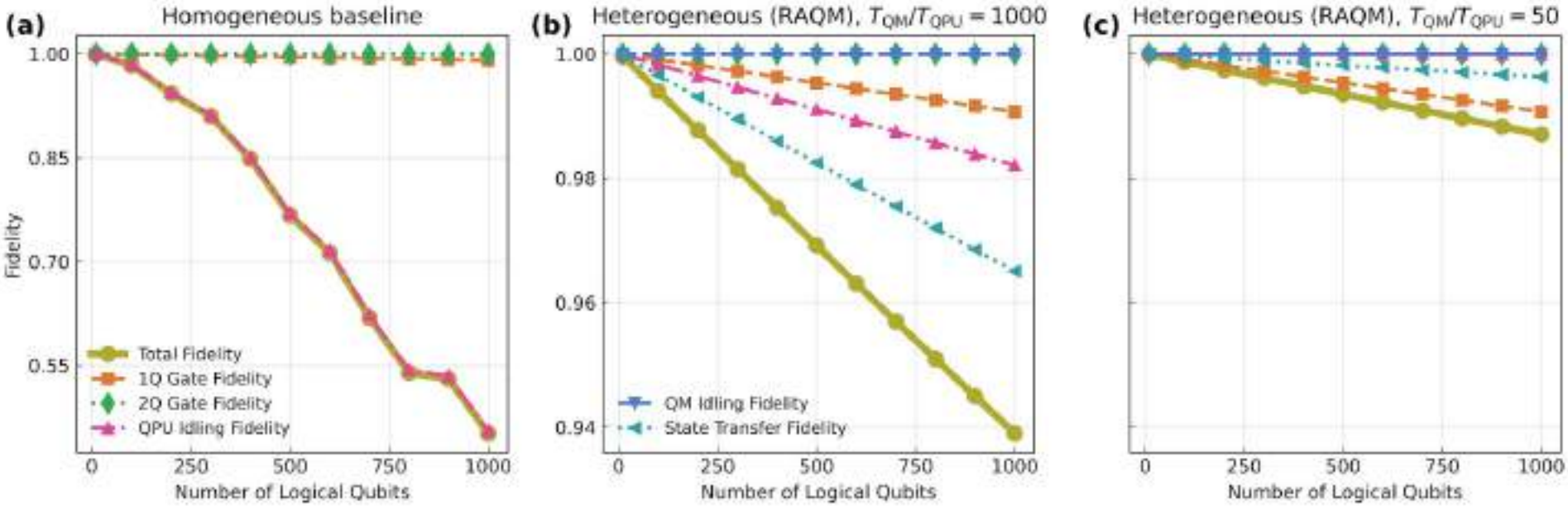
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Error budget analysis

Q-CHESS enables accurate error budget

Error budget is crucial for HW and software design



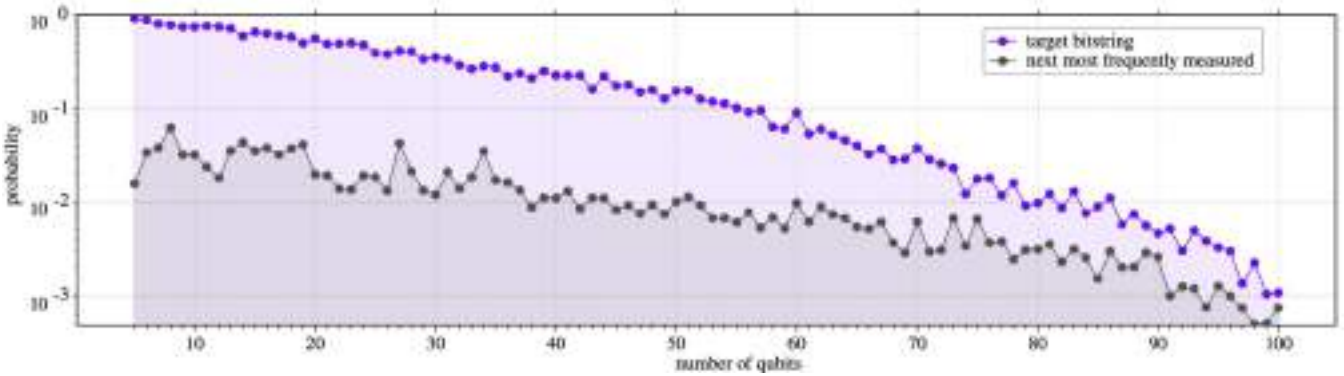
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Optimizing critical sub-routines

Specialized design for acceleration

Hardware aware design reduces dependency on hardware limitations

QFT Compilation Strategy	CX count QFT	CX count AQFT	CX depth QFT	CX depth AQFT
Low-CX LNN	$n^2 - n$	$d(2n - d - 1)$	$n^2 - 2n + 3$	$(2d - 1)n - d(d + 1) + 3$
Convolutional	$n^2 - n + 2$	$d(2n - d - 1) + 2$	$n^2 - 3n - 8$	$2(d - 1)n - d(d + 1) + 8$



- Eliminating the routing overhead for QFT for low connectivity devices.
- Allow the design of QFT modules with lower couplers/shuttling requirements.

arXiv:2608.05435

Decoder manager & orchestrator

Decoder manager and orchestrator

Decoders can’t be developed independently of the rest of the stack

- **Real time calibration**: Integrated our monitoring/calibration tools with the outer decoders, for real-time noise model and decoder parameters adjustments.
- **Code heterogeneity:** A decoding manager will handle the different decoding tasks.
- **Classical data flow**: reducing high decoding rates when possible.

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Thank you!



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